

OCR Further Mathematics
Statistics - Correlation
Question Paper
AS and A Level
Further Mathematics A - H235, H245



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The latest topic questions for the year 2024 / 2025+ exams.
To be used by all students preparing for OCR AS and A Level Further Mathematics.
Students of other boards may also find this useful.

- 1(a) The numbers of each of 9 items sold in two different supermarkets in a week are given in the following table.

Item	1	2	3	4	5	6	7	8	9
Supermarket A	17	28	41	43	62	69	75	93	115
Supermarket B	24	7	18	12	47	29	58	42	37

A researcher wants to test whether there is association between the numbers of these items sold in the two supermarkets.

However, it is known that the collection of data in Supermarket B was done inaccurately and each of the numbers in the corresponding row of the table could have been in error by as much as 2 items greater or 2 items fewer.

Explain why Spearman's rank correlation coefficient might be preferred to the use of Pearson's product-moment correlation coefficient in this context.

[2]

- (b) Carry out the test at the 5% significance level using Spearman's rank correlation coefficient.

[8]

- 2 A shopper estimates the cost, £X per item, of each of 12 items in a supermarket. The shopper's estimates are compared with the actual cost, £Y per item, of each item. The results are summarised as follows.

$$n = 12$$

$$\Sigma x = 399$$

$$\Sigma y = 623.88$$

$$\Sigma x^2 = 28\,127$$

$$\Sigma y^2 = 116\,509.0212$$

$$\Sigma xy = 45\,006.01$$

Test at the 1% significance level whether the shopper's estimates are positively correlated with the actual cost of the items.

[7]

- 3(a) A book collector compared the prices of some books, £x, when new in 1972 and the prices of copies of the same books, £y, on a second-hand website in 2018.

The results are shown in Table 1 and are summarised below the table.

Book	A	B	C	D	E	F	G	H	I	J	K	L
x	0.95	0.65	0.70	0.90	0.55	1.40	1.50	0.50	1.15	0.35	0.20	0.35
y	6.06	7.00	2.00	5.87	4.00	5.36	7.19	2.50	3.00	8.29	1.37	2.00

Table 1

$$n = 12, \Sigma x = 9.20, \Sigma y = 54.64, \Sigma x^2 = 8.9950, \Sigma y^2 = 310.4572, \Sigma xy = 46.0545$$

It is given that the value of Pearson's product-moment correlation coefficient for the data is 0.381, correct to 3 significant figures.

- (i) State what this information tells you about a scatter diagram illustrating the data.

[1]

- (ii) Test at the 5% significance level whether there is evidence of positive correlation between prices in 1972 and prices in 2018.

[5]

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- (b) The collector noticed that the second-hand copy of book J was unusually expensive and he decided to ignore the data for book J.

Calculate the value of Pearson's product-moment correlation coefficient for the other 11 books.

[2]



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4(a) Carry out an appropriate test at the 1% level.

[7]



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(b) Explain what is meant by the statement that the test carried out in the part above is a non-parametric test.

[1]

5(a) Five observations of bivariate data (x, y) are given in the table.

x	7	8	12	6	4
y	20	16	7	17	23

Find the value of Pearson's product-moment correlation coefficient.

[2]

(b) State what your answer to part (a) tells you about a scatter diagram representing the data.

[2]

(c) A new variable α is defined by $\alpha = 3x + 4$. Dee says "The value of Pearson's product-moment correlation coefficient between α and y will not be the same as the answer to part (a)."

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State with a reason whether you agree with Dee.

[1]

6(a) Five runners, A, B, C, D and E , take part in two different races.

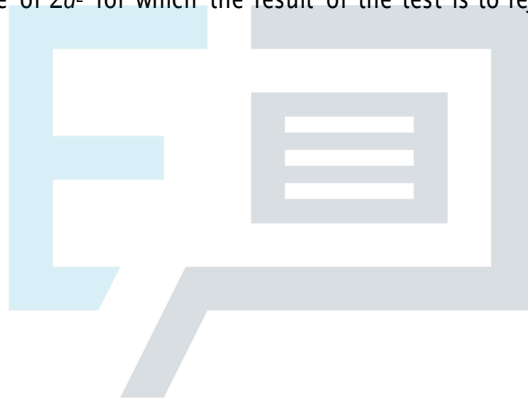
Spearman's rank correlation coefficient for the orders in which the runners finish is calculated and a test for positive agreement is carried out at the 5% significance level.

State suitable hypotheses for the test.

[1]

(b) Find the largest possible value of Σd^2 for which the result of the test is to reject the null hypothesis.

[3]



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- (c) In the first race, the order in which the five runners finished was: A, B, C, D, E . In the second race, three of the runners finished in the same positions as in the first race. The result of the test is to reject the null hypothesis.

Find a possible order for the runners to finish in the second race.

[3]

- 7(a) A set of bivariate data (X, Y) is summarised as follows.

$$n = 25, \Sigma x = 9.975, \Sigma y = 11.175, \Sigma x^2 = 5.725, \Sigma y^2 = 46.200, \Sigma xy = 11.575$$

Calculate the value of Pearson's product-moment correlation coefficient.

[1]

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(b) Calculate the equation of the regression line of y on x . [2]

(c) It is desired to know whether the regression line of y on x will provide a reliable estimate of y when $x = 0.75$.

State one reason for believing that the estimate will be reliable. [1]

(d) State what further information is needed in order to determine whether the estimate is reliable. [1]

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8(a) Sixteen candidates took an examination paper in mechanics and an examination paper in statistics.

For all sixteen candidates, the value of the product moment correlation coefficient r for the marks on the two papers was 0.701 correct to 3 significant figures.

Test whether there is evidence, at the 5% significance level, of association between the marks on the two papers.

[4]



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- (b) A teacher decided to omit the marks of the candidates who were in the top three places in mechanics and the candidates who were in the bottom three places in mechanics. The marks for the remaining 10 candidates can be summarised by

$$n = 10, \Sigma x = 750, \Sigma y = 690, \Sigma x^2 = 57\,690, \Sigma y^2 = 49\,676, \Sigma xy = 50\,829.$$

- (i) Calculate the value of r for these 10 candidates.

[2]



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- (ii) What do the two values of r , in part (a) and (b) (i), tell you about the scores of the sixteen candidates?

[1]

9(a) Shula calculates the value of Spearman's rank correlation coefficient r_s for 9 pairs of rankings.

Find the largest possible value of r_s that Shula can obtain that is less than 1.

[4]



(b) A set of bivariate data consists of 5 pairs of values. It is known that for this data the value of Spearman's rank correlation coefficient is -1 but the value of Pearson's product-moment correlation coefficient is not -1 .

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Sketch a possible scatter diagram illustrating the data.

[2]

- 10 The birth rate, x per thousand members of the population, and the life expectancy at birth, y years, in 14 randomly selected African countries are given in the table.

Country	x	y	Country	x	y
Benin	4.8	59.2	Mozambique	5.4	54.63
Cameroon	4.7	54.87	Nigeria	5.7	52.29
Congo	4.9	61.42	Senegal	5.1	65.81
Gambia	5.7	59.83	Somalia	6.5	54.88
Liberia	4.7	60.25	Sudan	4.4	63.08
Malawi	5.1	60.97	Uganda	5.8	57.25
Mauretania	4.6	62.77	Zambia	5.4	58.75

$$n = 14, \Sigma x = 72.8, \Sigma y = 826, \Sigma x^2 = 392.96, \Sigma y^2 = 48\,924.54, \Sigma xy = 4279.16$$

- (a) Calculate Pearson's product-moment correlation coefficient r for the data.

[2]

- (b) State what would be the effect on the value of r if the birth rate were given per hundred and not per thousand.

[1]

- (c) Explain what the sign of r tells you about the relationship between life expectancy and birth rate for these countries.

[1]

- (d) Test at the 5% significance level whether there is correlation between birth rate and life expectancy at birth in African countries.

[5]



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- (e) A researcher wants to estimate the life expectancy at birth in Zimbabwe, where the birth rate is 3.9 per thousand. Explain whether a reliable estimate could be obtained using the regression line of y on x for the given data.

[1]

- 11 The table shows the values of 5 observations of bivariate data (x , y).

x	4.6	5.9	6.5	7.8	8.3
y	15.6	10.8	10.4	10.1	9.7

$$n = 5, \Sigma x = 33.1, \Sigma y = 56.6, \Sigma x^2 = 227.95, \Sigma y^2 = 664.26, \Sigma xy = 362.37$$

- (a) Calculate Pearson's product-moment correlation coefficient r for the data.

[2]



- (b) State what this value of r tells you about a scatter diagram illustrating the data.

[1]

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- (c) Test at the 5% significance level whether there is association between x and y .

[5]

(d) State the value of Spearman's rank correlation coefficient r^s for the data.

[1]

(e) State whether r , r_s , or both or neither is changed when the values of x are replaced by

(i) $3x - 2$

[1]

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(ii) $\sqrt{x}$

[1]

- 12 Judith believes that mathematical ability and chess-playing ability are related. She asks 20 randomly chosen chess players, with known British Chess Federation (BCF) ratings X , to take a mathematics aptitude test, with scores Y . The results are summarised as follows.

$$n = 20, \Sigma x = 3600, \Sigma x^2 = 660\,500, \Sigma y = 1440, \Sigma y^2 = 105\,280, \Sigma xy = 260\,990$$

- (a) Calculate the value of Pearson's product-moment correlation coefficient r .

[2]



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- (b) State an assumption needed to be able to carry out a significance test on the value of r .

[1]

- (c) Assume now that the assumption in part (b) is valid. Test at the 5% significance level whether there is evidence that chess players with higher BCF ratings are better at mathematics. [4]



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- (d) There are two different grading systems for chess players, the BCF system and the international ELO system. The two sets of ratings are related by

$$\text{ELO rating} = 8 \times \text{BCF rating} + 650.$$

Magnus says that the experiment should have used ELO ratings instead of BCF ratings. Comment on Magnus's suggestion. [1]

13 At a wine-tasting competition, two judges give marks out of 100 to 7 wines as follows.

Wine	<i>A</i>	<i>B</i>	<i>C</i>	<i>D</i>	<i>E</i>	<i>F</i>	<i>G</i>
Judge I	86.3	87.5	87.6	88.8	89.4	89.9	90.5
Judge II	85.3	88.1	82.7	87.7	89.0	89.4	91.5

- (a) A spectator claims that there is a high level of agreement between the rank orders of the marks given by the two judges. Test the spectator's claim at the 1% significance level.

[7]



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- (b) A competitor ranks the wines in a random order. The value of Spearman's rank correlation coefficient between the competitor and Judge I is r_s .
- (i) Find the probability that $r_s = 1$. [2]



- (ii) Show that r_s cannot take the value $\frac{55}{56}$. [2]

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14 In a competition, entrants have to give ranks from 1 to 7 to each of seven resorts. The correct ranks for the resorts are decided by an expert.

- (a) One competitor chooses his ranks randomly. By considering all the possible rankings, find the probability that the value of Spearman's rank correlation coefficient r_s between the competitor's ranks and the expert's ranks is at least $\frac{27}{28}$. [7]



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- (b) Another competitor ranks the seven resorts. A significance test is carried out to test whether there is evidence that this competitor is merely guessing the rank order of the seven resorts. The critical region

is $r_s \geq \frac{27}{28}$. State the significance level of the test. [1]

- 15 The numbers of students taking A levels in three subjects at a school were classified by the year in which they entered the school as follows.

	Subject	Mathematics	English	Physics
Year of Entry	Year 7	17	16	7
	Year 12	13	2	5

The Head of the school carries out a significance test at the 10% level to test whether subjects taken are independent of year of entry.

- (a) Show that in carrying out the test it is necessary to combine columns.

[2]



- (b) Suggest a reason why it is more sensible to combine the columns for Mathematics and Physics than the columns for Physics and English.

[1]

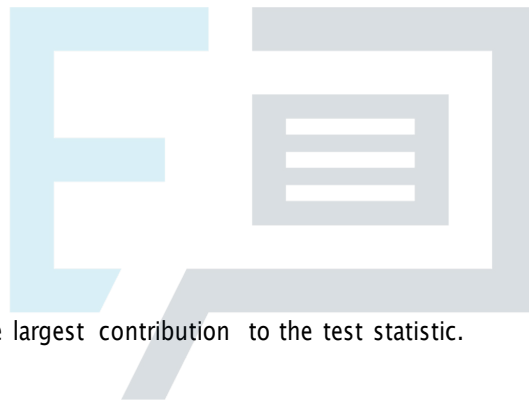
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- (c) Carry out the test.

[6]



(d) State which cell gives the largest contribution to the test statistic.

[1]

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(e) Interpret your answer to part (d).

[1]

16 The discrete random variable R has the distribution $Po(\lambda)$.

Use an algebraic method to find the range of values of λ for which the single most likely value of R is 7. [7]

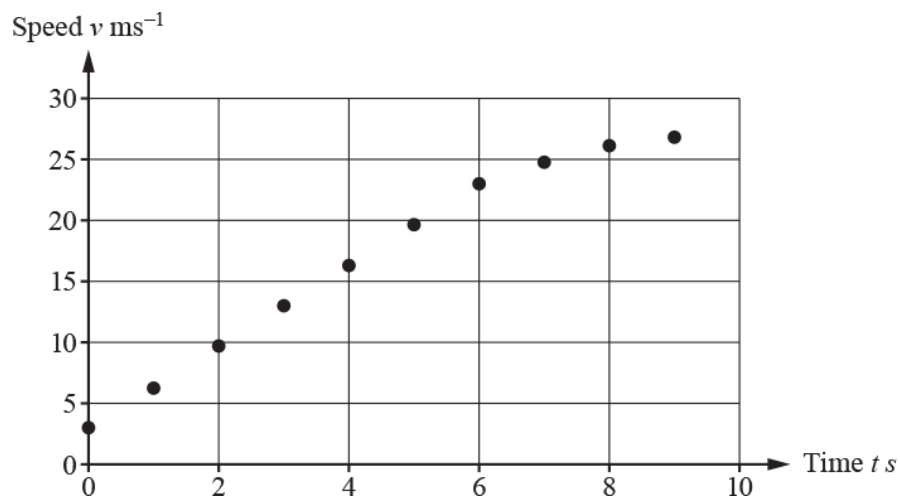


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17 The speed $v \text{ ms}^{-1}$ of a car at time t seconds after it starts to accelerate was measured at 1-second intervals. The results are shown in the following diagram.



(a) State whether t or v or neither is a controlled variable.

[1]

The value of the product moment correlation coefficient r for the data is 0.987 correct to 3 significant figures.

(b) The speed of the car is converted to miles per hour and the time to minutes. State the value of r for the converted data.

[1]

(c) State the value of Spearman's rank correlation coefficient r_s for the data.

[1]

(d) What information does r give about the data that is not given by r_s ?

[1]

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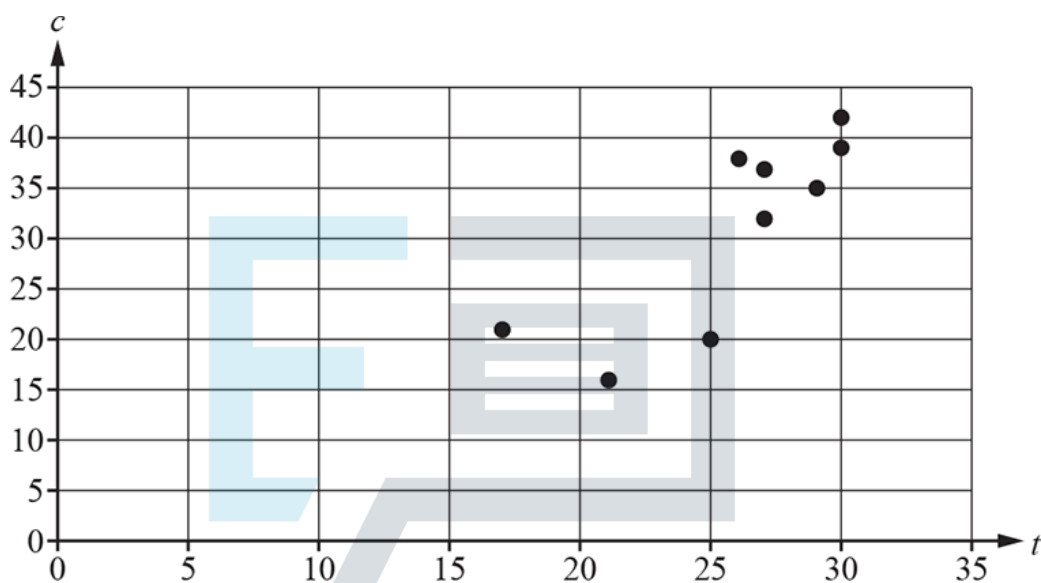
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- 18 A shop manager recorded the maximum daytime temperature T °C and the number C of ice creams sold on 9 summer days.

The results are given in the table and illustrated in the scatter diagram.

T	17	21	25	26	27	27	29	30	30
C	21	16	20	38	32	37	35	39	42



$$n = 9, \Sigma t = 232, \Sigma c = 280, \Sigma t^2 = 6130, \Sigma c^2 = 9444, \Sigma tc = 7489$$

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- (a) State, with a reason, whether one of the variables C or T is likely to be dependent upon the other. [1]

- (b) Calculate Pearson's product-moment correlation coefficient r for the data. [2]

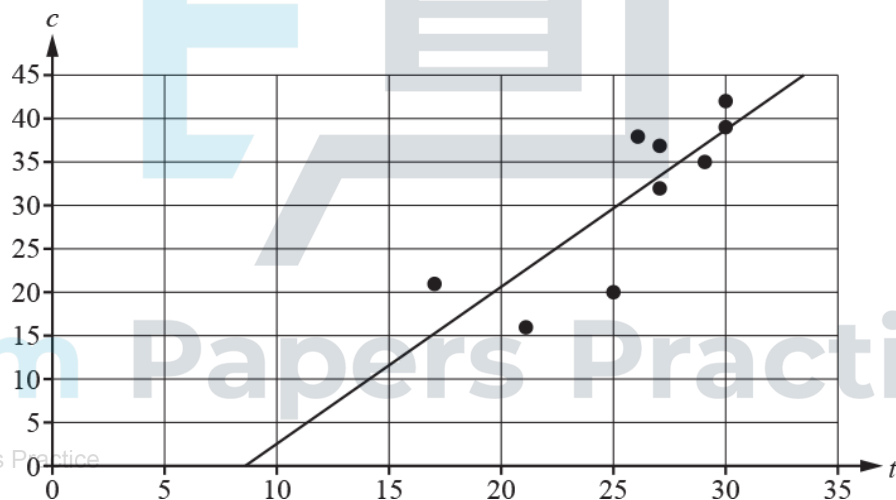
- (c) State with a reason what the value of r would have been if the temperature had been measured in °F rather than °C. [1]

(d) Calculate the equation of the least squares regression line of c on t .

[2]

(e) The regression line is drawn on the copy of the scatter diagram below. Use this diagram to explain what is meant by “least squares”.

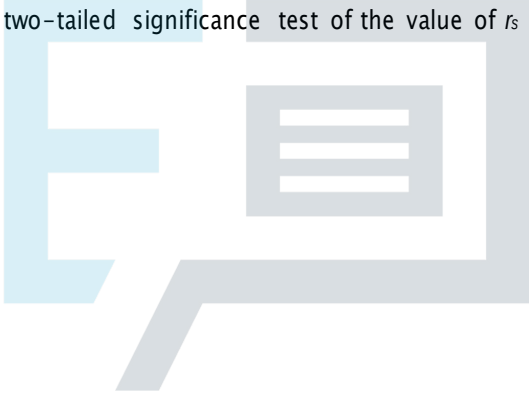
[2]



- 19 Arlosh, Sarah and Desi are investigating the ratings given to six different films by two critics.
- (a) Arlosh calculates Spearman's rank correlation coefficient r_s for the critics' ratings. He calculates that $\Sigma d^2 = 72$. Show that this value must be incorrect. [2]

- (b) Arlosh checks his working with Sarah, whose answer $r_s = \frac{29}{35}$ is correct. Find the correct value of Σd^2 . [2]

- (c) Carry out an appropriate two-tailed significance test of the value of r_s at the 5% significance level, stating your hypotheses clearly. [4]



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Each critic gives a score out of 100 to each film. Desi uses these scores to calculate Pearson's product-moment correlation coefficient. She carries out a two-tailed significance test of this value at the 5% significance level.

- (d) Explain with a reason whether you would expect the conclusion of Desi's test to be the same as the result of the test in part (c). [1]

- 20 The following table gives the mean per capita consumption of mozzarella cheese per annum, x pounds, and the number of civil engineering doctorates awarded, y , in the United States in each of 10 years.

x	9.3	9.7	9.7	9.7	9.9	10.2	10.5	11.0	10.6	10.6
y	480	501	540	552	547	622	655	701	712	708

source: www.tylervigen.com

- (a) Find the equation of the regression line of y on x . [2]

You are given that the product moment correlation coefficient is 0.959.

- (b) Explain whether this value would be different if x is measured in kilograms instead of pounds. [1]

It is desired to carry out a hypothesis test to investigate whether there is correlation between these two variables.

- (c) Assume that the data is a random sample of all years.

- (i) Carry out the test at the 10% significance level.

[6]

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- (ii) Explain whether your conclusion suggests that manufacturers of mozzarella cheese could increase consumption by sponsoring doctoral candidates in civil engineering. [1]

21 Two music critics, P and Q , give scores to seven concerts as follows.

Concert	1	2	3	4	5	6	7
Score by critic P	12	11	6	13	17	16	14
Score by critic Q	9	13	8	14	18	16	20

- (a) Calculate Spearman's rank correlation coefficient, r_s , for these scores. [4]

- (b) Without carrying out a hypothesis test, state what your answer tells you about the views of the two critics. [1]

- (i) The table shows the times, in minutes, spent by five students revising for a test, and the grades that they achieved in the test.

Student	Ann	Bill	Caz	Den	Ed
Time revising	0	60	35	100	45
Grade	C	D	E	B	A

Calculate Spearman's rank correlation coefficient.

[5]



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(ii) The table below shows the ranks given by two judges to four competitors.

Competitor	P	Q	R	S
Judge 1 rank	1	2	3	4
Judge 2 rank	3	2	1	4

Spearman's rank correlation coefficient for these ranks is denoted by r_s . With the same set of ranks for Judge 1, write down a different set of ranks for Judge 2 which gives the same value of r_s . There is no need to find the value of r_s .

[2]



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- 23 The table shows some of the values of the seasonally adjusted Unemployment Rate (UR), $x\%$, and the Consumer Price Index (CPI), $y\%$, in the United Kingdom from April 2008 to July 2010.

Date	April 2008	July 2008	October 2008	January 2009	April 2009	July 2009	October 2009	January 2010	April 2010	July 2010
UR, $x\%$	5.2	5.7	6.1	6.8	7.5	7.8	7.8	7.9	7.8	7.7
CPI, $y\%$	3.0	4.4	4.5	3.0	2.3	1.8	1.5	3.5	3.7	3.1

These data are summarised below.

$$n = 10 \qquad \Sigma x = 70.3 \qquad \Sigma x^2 = 503.45 \qquad \Sigma y = 30.8 \qquad \Sigma y^2 = 103.94 \qquad \Sigma xy = 211.9$$

- (i) Calculate the product moment correlation coefficient, r , for the data, showing that $-0.6 < r < -0.5$.

[3]

- (ii) Karen says “The negative value of r shows that when the Unemployment Rate increases, it causes the Consumer Price Index to decrease.” Give a criticism of this statement.

[1]

(iii)

- (a) Calculate the equation of the regression line of x on y .

[3]

- (b) Use your equation to estimate the value of the Unemployment Rate in a month when the Consumer Price Index is 4.0%.

[2]

- 24 The Gross Domestic Product per Capita (GDP), x dollars, and the Infant Mortality Rate per thousand (IMR), y , of 6 African countries were recorded and summarised as follows.

$$n = 6$$

$$\Sigma x = 7000$$

$$\Sigma x^2 = 8\,700\,000$$

$$\Sigma y = 456$$

$$\Sigma y^2 = 36\,262$$

$$\Sigma xy = 509\,900$$

- (i) Calculate the equation of the regression line of y on x for these 6 countries.

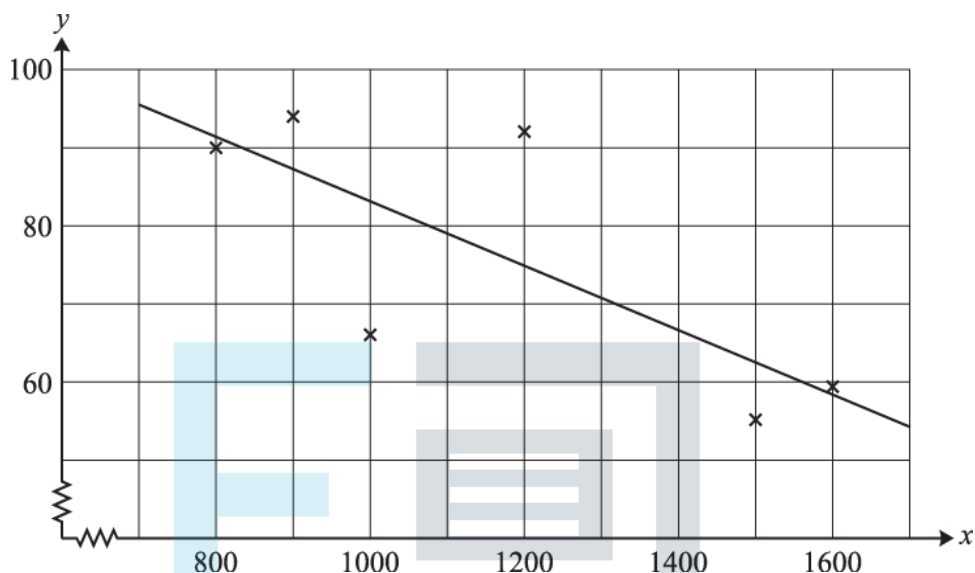
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[4]

The original data were plotted on a scatter diagram and the regression line of y on x was drawn, as shown below.



- (ii) The GDP for another country, Tanzania, is 1300 dollars. Use the regression line in the diagram to estimate the IMR of Tanzania.

[1]

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- (iii) The GDP for Nigeria is 2400 dollars. Give two reasons why the regression line is unlikely to give a reliable estimate for the IMR for Nigeria.

[2]

- (iv) The actual value of the IMR for Tanzania is 96. The data for Tanzania ($x = 1300$, $y = 96$) is now included with the original 6 countries. Calculate the value of the product moment correlation coefficient, r , for all 7 countries.

[4]

- (v) The IMR is now redefined as the infant mortality rate per hundred instead of per thousand, and the value of r is recalculated for all 7 countries. Without calculation state what effect, if any, this would have on the value of r found in part (iv).

[1]

- (i) Two judges rank n competitors, where n is an even number. Judge 2 reverses each consecutive pair of ranks given by Judge 1, as shown.

Competitor	C_1	C_2	C_3	C_4	C_5	C_6		C_{n-1}	C_n
Judge 1 rank	1	2	3	4	5	6		$n - 1$	n
Judge 2 rank	2	1	4	3	6	5		n	$n - 1$

Given that the value of Spearman's coefficient of rank correlation is $\frac{63}{65}$, find n .



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[4]

- (ii) An experiment produced some data from a bivariate distribution. The product moment correlation coefficient is denoted by r , and Spearman's rank correlation coefficient is denoted by r_s .

(a) Explain whether the statement

$$r = 1 \Rightarrow r_s = 1$$

is true or false.

[1]

(b) Use a diagram to explain whether the statement

$$r \neq 1 \Rightarrow r_s \neq 1$$

is true or false.

[2]

- (i) The table shows the amount, x , in hundreds of pounds, spent on heating and the number of absences, y , at a factory during each month in 2014.

Amount, x , spent on heating (£ hundreds)	21	23	19	15	14	5	2	10	9	20	18	23
Number of absences, y	23	25	18	18	12	10	4	9	11	15	20	26

$$n = 12$$

$$\Sigma x = 179$$

$$\Sigma x^2 = 3215$$

$$\Sigma y = 191$$

$$\Sigma y^2 = 3565$$

$$\Sigma xy = 3343$$

- (a) Calculate r , the product moment correlation coefficient, showing that $r > 0.92$.



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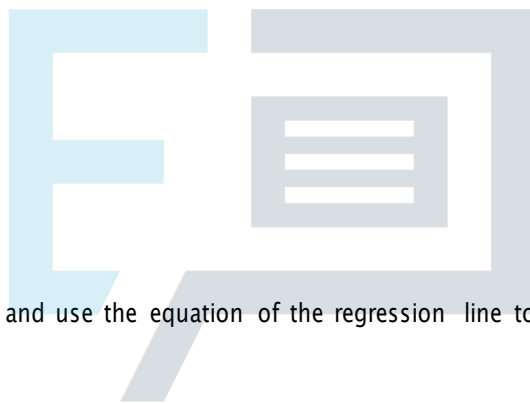
[3]

- (b) A manager says, 'The value of r shows that spending more money on heating causes more absences, so we should spend less on heating.' Comment on this claim.

[1]

- (ii) The months in 2014 were numbered 1, 2, 3, ..., 12. The output, z , in suitable units was recorded along with the month number, n , for each month in 2014. The equation of the regression line of z on n was found to be $z = 0.6n + 17$.

- (a) Use this equation to explain whether output generally increased or decreased over these months. [1]



- (b) Find the mean of n and use the equation of the regression line to calculate the mean of z .

[3]

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- (c) Hence calculate the total output in 2014.

[2]

27 In this question the product moment correlation coefficient is denoted by r and Spearman's rank correlation coefficient is denoted by r_s .

(i) The scatter diagram in Fig. 1 shows the results of an experiment involving some bivariate data.



Fig. 1

Write down the value of r_s for these data.

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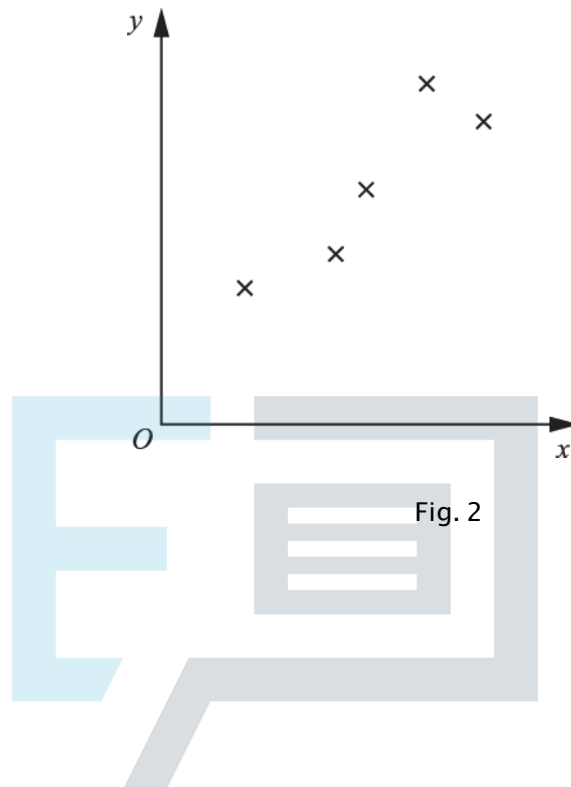
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[1]

(ii) On the diagram in the Answer Booklet, draw five points such that $r_s = 1$ and $r \neq 1$.

[2]

(iii) The scatter diagram in Fig. 2 shows the results of another experiment involving 5 items of bivariate data.



Calculate the value of r_s .

[5]

- 28 For the top 6 clubs in the 2010/11 season of the English Premier League, the table shows the annual salary, £ x million, of the highest paid player and the number of points scored, y .

Club	Manchester United	Manchester City	Chelsea	Arsenal	Tottenham	Liverpool
x	5.6	7.4	6.5	4.1	3.6	6.5
y	80	71	71	68	62	58

$$n = 6 \qquad \Sigma x = 33.7 \qquad \Sigma x^2 = 200.39 \qquad \Sigma y = 410 \qquad \Sigma y^2 = 28\,314 \qquad \Sigma xy = 2313.9$$

- (i) Use a suitable formula to calculate the product moment correlation coefficient, r , between x and y , showing that $0 < r < 0.2$.

[3]

- (ii) State what this value of r shows in this context.

[1]

- (iii) A fan suggests that the data should be used to draw a regression line in order to estimate the number of points that would be scored by another Premier League club, whose highest paid player's salary is £1.7 million. Give two reasons why such an estimate would be unlikely to be reliable.

[2]

- 29 An expert tested the quality of the wines produced by a vineyard in 9 particular years. He placed them in the following order, starting with the best.

1980 1983 1981 1982 1984 1985 1987 1986 1988

- (i) Calculate Spearman's rank correlation coefficient, r_s , between the year of production and the quality of these wines. The years should be ranked from the earliest (1) to the latest (9).

[5]

- (ii) State what this value of r_s shows in this context.

[1]

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- 30 Tariq collected information about typical prices, £y million, of four-bedroomed houses at varying distances, x miles, from a large city. He chose houses at 10-mile intervals from the city. His results are shown below.

x	10	20	30	40	50	60	70	80
y	1.2	1.4	1.2	0.9	0.8	0.5	0.5	0.3

$$n = 8 \qquad \Sigma x = 360 \qquad \Sigma x^2 = 20\,400 \qquad \Sigma y = 6.8 \qquad \Sigma y^2 = 6.88 \qquad \Sigma xy = 241$$

- (i) Use an appropriate formula to calculate the product moment correlation coefficient, r , showing that $-1.0 < r < -0.9$.

[3]

- (ii) State what this value of r shows in this context.

[1]

- (iii) Tariq decides to recalculate the value of r with the house prices measured in hundreds of thousands of pounds, instead of millions of pounds. State what effect, if any, this will have on the value of r .

[1]

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- (iv) Calculate the equation of the regression line of y on x .

[3]

- (v) Explain why the regression line of y on x , rather than x on y , should be used for estimating a value of x from a given value of y .

[1]

31 Fiona and James collected the results for six hockey teams at the end of the season. They then carried out various calculations using Spearman's rank correlation coefficient, r_s .

- (i) Fiona calculated the value of r_s between the number of goals scored FOR each team and the number of goals scored AGAINST each team. She found that $r_s = -1$. Complete the table in the answer book showing the ranks.

Team	A	B	C	D	E	F
Number of goals FOR (rank)	1	2	3	4	5	6
Number of goals AGAINST (rank)						

[1]

- (ii) James calculated the value of r_s between the number of goals scored and the number of points gained by the 6 teams. He found the value of r_s to be 1. He then decided to include the results of another two teams in the calculation of r_s . The table shows the ranks for these two teams.

Team	G	H
Number of goals scored (rank)	7	8
Number of points gained (rank)	8	7

Calculate the value of r_s for all 8 teams.

[4]

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END OF QUESTION PAPER